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Effects of Al contents on aging hardening behaviors of Fe-Mn-Al-Cu low-density steels with ultra-high specific hardness

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ABSTRACT

Low-density steels have garnered significant attention due to the demand for lightweight structural materials in aerospace and automobile industries. However, manufacturing low-density steels with different hardness levels poses challenges in composition design, processing optimization, and the application of hardening techniques across various alloy systems. In this work, Fe-35Mn-(15, 17, 19)Al-5Cu (at.%) low-density steels with an ultra-wide flexible hardness ranging from 253.4 HV to 965.6 HV through simple aging treatments were developed, attributing to the continuous precipitation of the ultra-hard β-Mn phase. The densities of the Fe-Mn-Al-Cu steels were regulated by varying the Al contents, and the effects of Al contents on the aging hardening behaviors of Fe-Mn-Al-Cu low-density steels were systematically investigated. The results show that the increase of Al content led to the decrease in precipitation kinetics of β-Mn. Moreover, increasing Al content slightly enhanced the intrinsic hardness of β-Mn. However, it also led to the decrease of FCC Cu-rich phase between β-Mn interfaces and the decrease in the fracture toughness of the ultra-hard alloys. This resulted in the decrease in the final hardening effect of Fe-Mn-Al-Cu low-density steels with increasing Al content. Due to the advantages of ultra-high hardness (965.6 HV) and low density, our Fe-35.2Mn-15Al-5.2Cu low-density steel achieved an ultra-high specific hardness of 141 HV·cm^3/g.

1. Introduction

In recent years, low-density steel, also known as lightweight steel, has garnered widespread interest due to the demand for lightweight structural materials [1–5]. Low-density steels refer to a class of steels with a lower density compared to traditional steels, typically achieved through alloying design [6–8]. Ham and Cairns initially developed a cost-effective Fe-34Mn-10Al-0.76C low-density steel to replace Cr—Ni stainless steel in mildly corrosive environments [9]. In this steel, the Al and Mn were used as substitutes for Cr and Ni, respectively, to enhance its corrosion resistance and high-temperature oxidation resistance. Subsequent research on low-density steels primarily focused on reducing material density without significantly compromising mechanical properties, in order to meet the demands for high-strength lightweight metallic materials in aerospace, automotive, and other industries [10–12]. It is estimated that reducing the density of steels by just 10 % could maintain or even significantly enhance its superiority and competitiveness in industrial applications owing to their high specific strength and specific hardness [13,14]. The specific hardness and specific strength of low-density steels are critical in industrial applications, particularly when used as structural materials. Enhancing hardness in low-density steels offers significant benefits, especially in applications where wear resistance and long service life are critical [15–18]. However, further increasing the hardness of low-density steels remains a formidable challenge.

Low-density steels primarily consist of binary Fe-Al, Fe-Mn-Al, and Fe-Mn-Si-Al alloys [7,8,19]. Among them, Fe-Mn-Al alloys are highly promising in engineering applications thanks to their outstanding mechanical properties and reduced specific weight. Additionally, Fe-Mn-Al alloys have been reported to possess several appealing advantages, such as high strength and toughness at room and low temperatures [20,21], excellent fatigue properties [22], and good oxidation resistance at elevated temperatures [23]. Notably, Fe-Mn-Al low-density steels exhibit remarkable aging hardening capabilities, allowing for a

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Table 1
Chemical compositions of the 15Al, 17Al and 19Al alloys.

Alloys	Chemical compositions (at. %)			
	Fe	Mn	Al	Cu
15Al	44.6	35.2	15.0	5.2
17Al	43.2	34.8	17.0	5.0
19Al	41.6	34.8	18.6	5.0

significant increase in hardness through heat treatments. This is primarily attributed to the precipitation of the ultra-hard β-Mn during the aging process [24–26]. However, previous research indicates that the hardening effect of β-Mn precipitated in Fe-Mn-Al low-density steels is still limited, with the maximum hardness typically falling below 800 HV [24,25]. Furthermore, the slow precipitation rate of β-Mn during aging would extend the aging time for achieving high hardness. For instance, the maximum hardness value achieved in Fe-31Mn-11Al-0.89C steel was 750 HV after aging at 550 °C for over 1000 min [24]. This limitation is not only due to the slow precipitation rate of β-Mn but also related to its insufficient formation amount. Therefore, further optimizing the precipitation behavior to promote the precipitation of β-Mn in Fe-Mn-Al low-density steels during the aging process is crucial for further enhancing the hardness of Fe-Mn-Al low-density steels. This could further broaden the application range of Fe-Mn-Al low-density steels and save production costs and time.

We have previously developed an Fe-Mn-Al-Cu alloy with rapid aging hardening response and ultra-high hardness [15]. The addition of Cu not only promoted the precipitation kinetics of β-Mn, but also enhanced the hardness of Fe-Mn-Al-Cu low-density steel. Similarly, Al plays an important role not only in the density and mechanical properties of low-density steels, but also possibly in the precipitation and hardening behavior of β-Mn in the Fe-Mn-Al-Cu low-density steels during aging treatments. In this work, we investigated the effects of Al contents on the aging hardening behaviors of Fe-35Mn-xAl-5Cu (at.%, x = 15, 17, 19) low-density steels based on considering the dual criteria of low density and high hardness. Firstly, high Mn content of 35 at.% was chosen to promote the precipitation kinetics of hard β-Mn, mainly responsible for the achieved ultrahigh and flexible hardness. In addition, the addition of different Al contents is due to the following considerations: 1) Al is the primary element responsible for reducing the density in Fe-Mn-Al-Cu low-density steels. The addition of 1 wt.% Al reduces the density by approximately 0.1 g/cm^3 [21,27]. 2) The effect of Al on the precipitation behavior of β-Mn during the aging process of Fe-Mn-Al-Cu low-density steels has yet to be thoroughly explored. Through aging treatments between 525 °C and 650 °C of no more than 10 h, we achieved an exceptionally wide and adjustable range of hardness in these Fe-Mn-Al-Cu low-density steels, spanning from 253.4 HV to 965.6 HV. This was primarily attributed to the rapid precipitation of the hard β-Mn phase. In addition, an ultra-high specific hardness of 141 HV·cm^3/g were achieved in the Fe-35.2Mn-15Al-5.2Cu low-density steel. These findings pave a new way for developing low-density steels with ultra-high and flexible hardness within a wide range.

2. Materials and methods

2.1. Alloy preparation

The alloy ingots with the nominal compositions of Fe-35Mn-xAl-5Cu (at.%, x = 15, 17, 19) were produced through melting raw materials with high purity (>99.9 %) using a vacuum induction furnace under an argon atmosphere. The alloys are referred to as 15Al, 17Al, and 19Al according to their Al contents, respectively. The chemical compositions measured via Energy Dispersive Spectroscopy (EDS) are detailed in Table 1. The densities of 15Al, 17Al and 19Al alloys were measured by Archimedes method as 6.85, 6.76, 6.66 g·cm^{-3}. The ingots were homogenized at 1200 °C for 24 h, then hot-rolled into plates with a thickness of 4 mm. Subsequently, these plates were cut into 10 × 5 × 2 mm^3 samples using electrical discharge machining for Vickers hardness measurements and microstructure characterizations. The samples were solution treated at 1100 °C for 10 min under an argon atmosphere, followed by oil quenching. The solution-treated samples were then aged at temperatures between 525 °C and 650 °C for 1, 3, 6 and 10 h, followed by water quenching. Samples for Vickers hardness tests, X-Ray Diffraction (XRD), scanning electron microscopy (SEM), and electron backscattered diffraction (EBSD) characterizations were prepared by electrolytically polishing using a mixture of 10 % perchloric acid and 90 % ethanol. Samples for transmission electron microscope (TEM) observation were first mechanically ground to thin foils with thicknesses of about 90 μm, then punched into foils with diameters of 3 mm. Finally, the TEM foils were ion thinned.

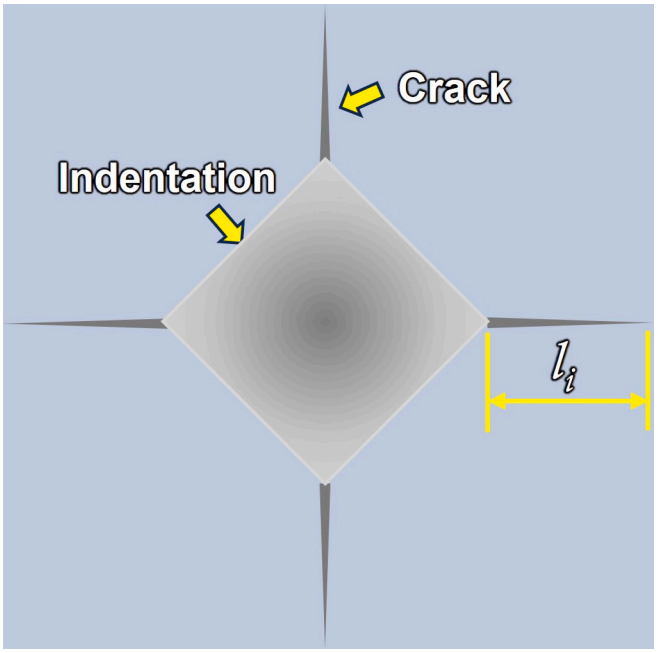


Fig. 1. Schematic diagram demonstrating the approach to evaluating fracture toughness using indentation techniques.

2.2. Hardness measurements

The Vickers hardness of the samples was measured using an HVS-1000 digital microhardness tester with a load of 9.8 N for 15 s. Each sample was repeatedly tested 5 times, and the hardness values were given by the average of 5 measurements. Samples for nanoindentation tests were mechanically polished to prevent the Cu-rich phase from flaking due to electrolytic polishing. Nanoindentation tests were conducted using an Agilent G200 Nanoindenter. The indentation depth was 200 nm, and each test point was spaced 5 μm or more apart.

The fracture toughness (K_{IC}, MPa·m$^{1/2}$) values of the Fe-Mn-Al-Cu low-density steels with peak hardness values (HV = 965.6 for 15Al, HV = 924.9 for 17Al, and HV = 917.8 for 19Al) were measured using the indentation technique [28]. The fracture toughness of the samples was evaluated based on the crack propagation. K_{IC} was calculated using Eq. (1):

$$K_{IC} = 0.016\left(\frac{E}{HV}\right)^{1/2}\frac{P}{c^{2/3}} \quad (1)$$

where E is the elastic modulus (164 GPa for 15Al, 175 GPa for 17Al and 181 GPa for 19Al measured through compressive stress-strain curves), HV is the Vickers hardness, P is the indentation load (9.8 N), and c is the average crack length ($c = 0.25\sum_{i=1}^{4} l_i$, in which l_i is the radical crack

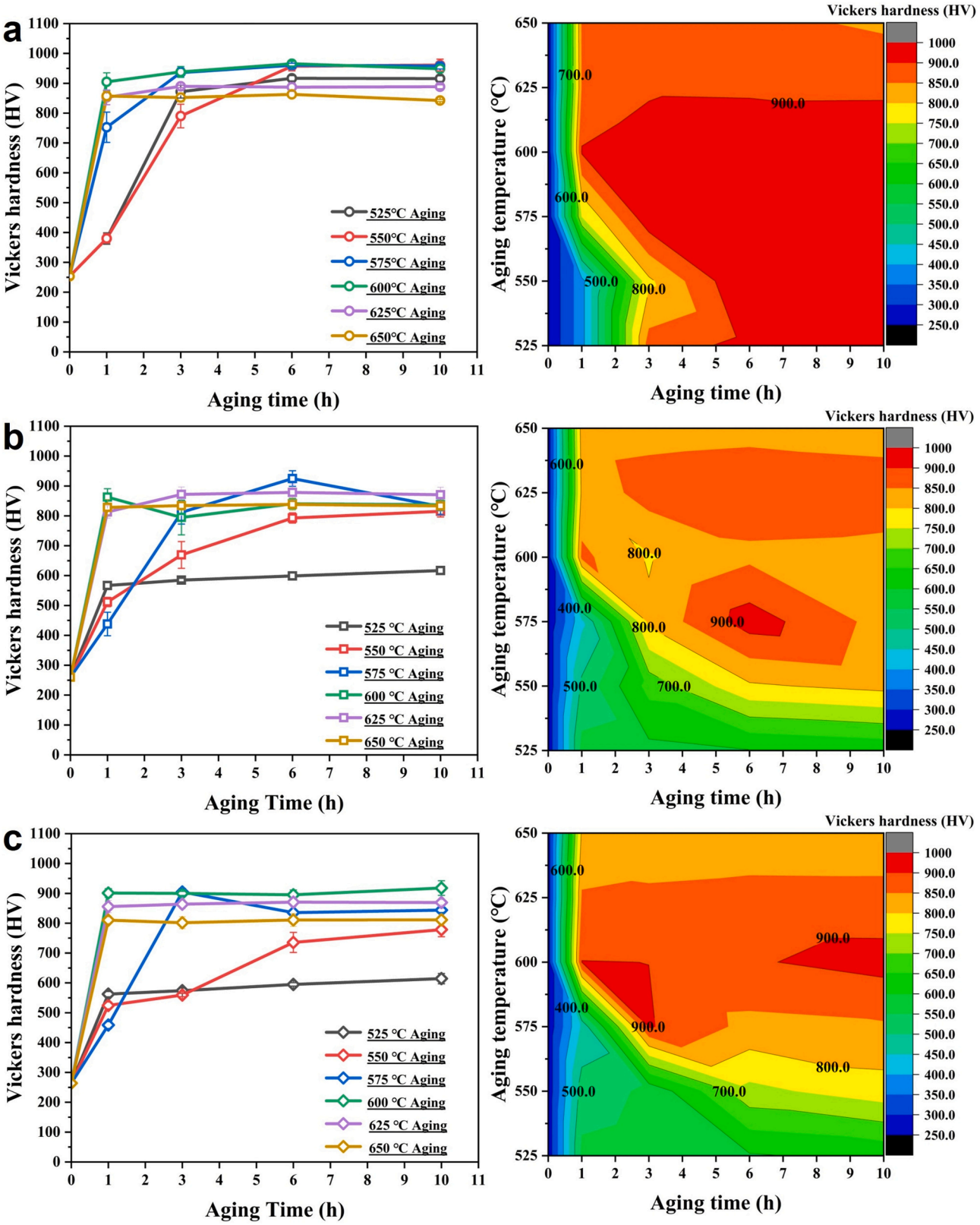


Fig. 2. Evolutions of Vickers hardness and corresponding hardness contour plots of the (a) 15Al, (b) 17Al, and (c) 19Al alloys after different aging treatments.

length starting from the indentation corners), as schematically illustrated in Fig. 1.

2.3. Microstructure characterizations

Phase compositions of the alloys were identified using X-ray diffraction (XRD) through a D8 Discover X-ray diffractometer. The secondary electronic (SE) and EDS analyses were performed on a ZEISS

Table 2
Vickers hardness values of the 15Al alloys after different aging treatments.

Vickers hardness (HV)	Solution-treated	1 h	3 h	6 h	10 h
525 °C aging	253.4 ± 2.1	379.8 ± 19.1	872.0 ± 21.2	916.8 ± 8.7	915.7 ± 9.5
550 °C aging		380.2 ± 15.5	790.3 ± 23.5	957.1 ± 15.5	961.1 ± 19.6
575 °C aging		752.6 ± 50.7	935.5 ± 15.8	960.3 ± 6.8	957.8 ± 10.1
600 °C aging		904.9 ± 30.2	938.2 ± 18.0	965.6 ± 7.6	948.6 ± 12.8
625 °C aging		852.8 ± 24.6	889.9 ± 4.3	887.2 ± 14.1	888.2 ± 6.6
650 °C aging		857.1 ± 15.4	852.3 ± 13.8	862.8 ± 11.3	842.3 ± 3.7

Table 3
Vickers hardness values of the 17Al alloys after different aging treatments.

Vickers hardness (HV)	Solution-treated	1 h	3 h	6 h	10 h
525 °C	259.8 ± 1.5	566.9 ± 9.7	585.0 ± 6.1	598.9 ± 10.4	616.8 ± 12.3
550 °C aging		511.6 ± 15.6	669.2 ± 24.9	792.9 ± 16.9	815.3 ± 19.1
575 °C aging		438.1 ± 39.3	811.7 ± 39.3	924.9 ± 25.8	831.3 ± 27.4
600 °C aging		862.7 ± 28.0	794.6 ± 58.1	840.4 ± 18.5	837.3 ± 15.3
625 °C aging		813.0 ± 20.8	872.0 ± 24.4	878.4 ± 28.3	870.5 ± 24.9
650 °C aging		828.4 ± 26.6	834.2 ± 14.0	838.3 ± 20.7	833.2 ± 18.7

Table 4
Vickers hardness values of the 19Al alloys after different aging treatments.

Vickers hardness (HV)	Solution-treated	1 h	3 h	6 h	10 h
525 °C aging	263.8 ± 1.1	562.3 ± 8.9	574.1 ± 8.6	594.9 ± 5.7	614.4 ± 17.3
550 °C aging		524.5 ± 11.6	559.4 ± 5.2	778.5 ± 23.7	735.5 ± 23.5
575 °C aging		458.7 ± 11.5	904.5 ± 12.0	835.4 ± 30.9	843.7 ± 7.9
600 °C aging		901.2 ± 14.6	899.8 ± 6.8	895.3 ± 16.2	917.8 ± 24.5
625 °C aging		855.6 ± 20.8	863.9 ± 15.8	870.7 ± 16.1	869.5 ± 8.6
650 °C aging		810.1 ± 10.4	801.5 ± 20.3	810.9 ± 21.7	811.2 ± 19.5

Sigma 300 SEM operated at 5 kV. The EBSD was conducted on a GeminiSEM 300 SEM, and the scanning step size was 1.5 μm for low magnification and 0.05 μm for high magnification observations. The TEM observations were performed on an FEI Talos F200X microscope equipped with an EDS detector, operated at 200 kV.

3. Results

3.1. Aging hardening behaviors of Fe-Mn-Al-Cu low-density steels

Fig. 2 shows the aging hardening behaviors of the 15Al, 17Al, and 19Al alloys after different aging treatments, with corresponding average hardness values listed in Tables 2, 3, and 4, respectively. The hardness of the solution-treated alloys increased with rising Al content, attributed to the solid-solution strengthening effect [29]. After aging treatments, unusual rapid aging hardening behaviors were observed in all three alloys. Specifically, the achieved hardness ranges were 253.4 HV to 965.6 HV for 15Al, 259.8 HV to 924.9 HV for 17Al, and 263.8 HV to 917.8 HV for 19Al. The hardness increased slightly in the 15Al alloy after aging at 525 °C and 550 °C for 1 h. While a significant increase in hardness was observed when extending the aging time to 6 h, reaching the peak values over 900 HV. At temperatures of 575 °C and above, the aging hardening response was significantly accelerated. The hardness showed a substantial increase after just 1 h of aging and then slightly increased to reach the peak levels after aging for 3 h. The highest Vickers hardness of 965.6 HV was achieved when the 15Al alloy was aged at 600 °C for 6 h. After reaching the peak hardness, further extension of the aging time would lead to a slight decrease in hardness. The obtained hardness values when the 15Al alloy was aged at 625 °C and 650 °C were stabilized between 800 HV and 900 HV with prolonged aging time.

The aging hardening responses of the 17Al and 19Al alloys were similar to that of the 15Al alloy. Their hardness increased gradually during low-temperature aging at 525 °C and 550 °C, but they failed to reach the peak hardness. During aging at 575 °C and above, the rate of hardness increase was accelerated, while the maximum hardness value decreased. At the aging temperatures from 525 °C to 575 °C, the aging hardening response of the alloys significantly slowed down with increasing Al content. In addition, the peak hardness values after aging at higher temperatures decreased slightly with increasing Al content, with 924.9 HV for the 17Al alloy after aging at 575 °C for 6 h and 917.8 HV for the 19Al alloy after aging at 600 °C for 10 h. According to the hardness contour plots, the optimal aging temperature range for achieving ultra-high hardness for 15Al alloy was mainly in the range of 525 °C - 600 °C. However, the optimal aging temperature range to achieve such ultra-high hardness increased with increasing Al content, primarily in the range of 550 °C - 600 °C for 17Al alloy and 575 °C - 625 °C for 19Al alloy.

3.2. Evolutions of constituent phases after different aging treatments

The EBSD results depicted in Fig. 3 demonstrate that a single α (Body-centered cubic, BCC) phase formed in the solution-treated Fe-Mn-Al-Cu low-density steels. To clarify the microstructural origin of the rapid aging hardening behaviors of Fe-Mn-Al-Cu low-density steels, EBSD characterization was performed on these alloys subjected to different aging treatments. Fig. 4 presents the EBSD results of the 15Al alloy after different aging treatments. After aging at 525 °C for 3 h, it is evident that a significant amount of β-Mn had precipitated (85.8 %), with only a small amount of α matrix remaining untransformed. The formation of β-Mn led to the increase in hardness from 253.4 HV at the solid-solution state to 872.0 HV, indicating the strong hardening effect of the β-Mn phase. After aging at 550 °C for 6 h, the α matrix had almost entirely transformed into β-Mn. Furthermore, a new fine BCC phase was present within the β-Mn. Following aging at 650 °C for 3 h, the untransformed α regions were no longer observed, and a new FCC phase was distributed in β-Mn.

For the 17Al and 19Al alloys, the precipitation rate of β-Mn was noticeably slowed down when compared to the 15Al alloy. Fig. 5 shows that after aging the 17Al alloy at 525 °C for 10 h, only a small amount of β-Mn (20.5 %) formed along the α grain boundaries and within the α grains. In addition, some small α-Mn particles presented in the α matrix, conistent with our previous study [15]. In samples aged at 575 °C for 6 h and at 650 °C for 10 h, the α matrix had almost completely transformed into β-Mn. Besides, a small amount of new precipitates with BCC (Red highlighted) and FCC (Pink highlighted) crystalline structures were present within the β-Mn. Fig. 6 indicates that after aging the 19Al alloy at 525 °C for 10 h, only a small amount of β-Mn precipitated (12.1 %), resulting in a minimal increase in hardness. It is noteworthy that in the sample aged at a high temperature of 650 °C for 6 h, the α matrix had almost completely transformed into β-Mn, and a great amount of new BCC phase was observed within the β-Mn.

Based on the EBSD results, it can be inferred that the obtained ultra-high hardness originates from the extensive formation of β-Mn. Fig. 7

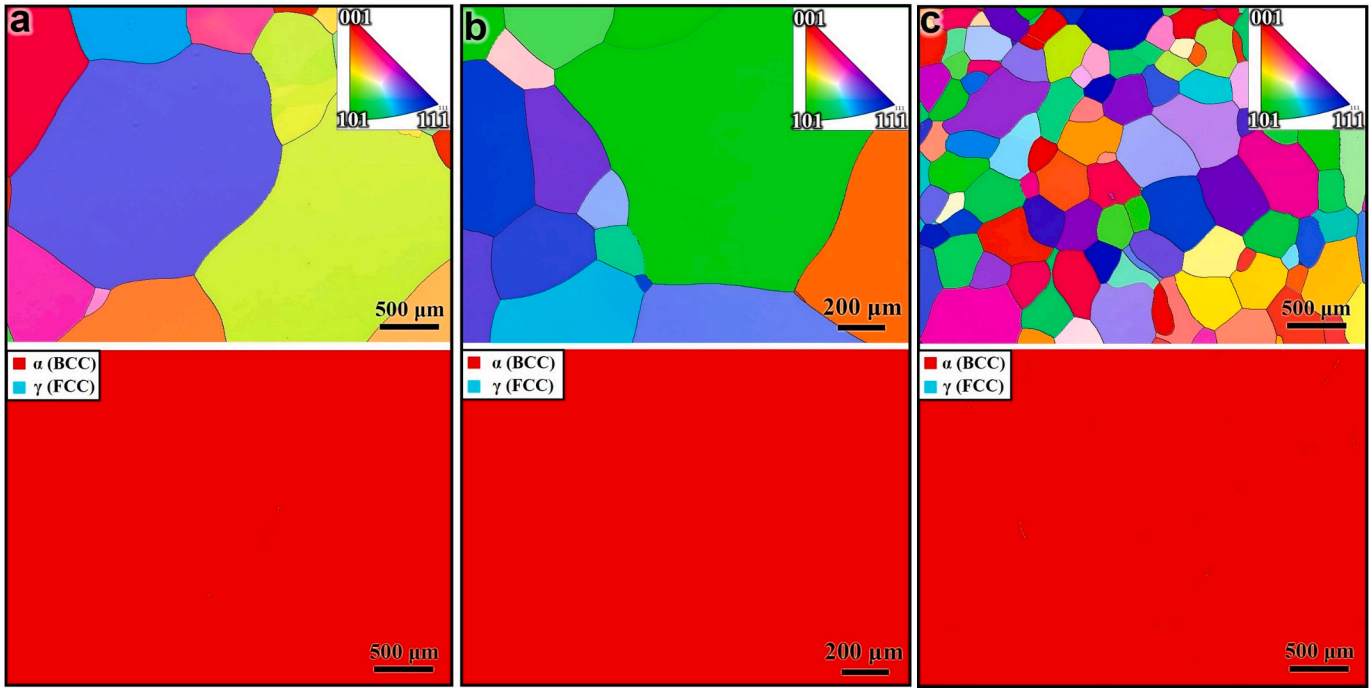


Fig. 3. EBSD inverse pole figures and corresponding phase maps of the solution-treated (a) 15Al, (b) 17Al, and (c) 19Al alloys.

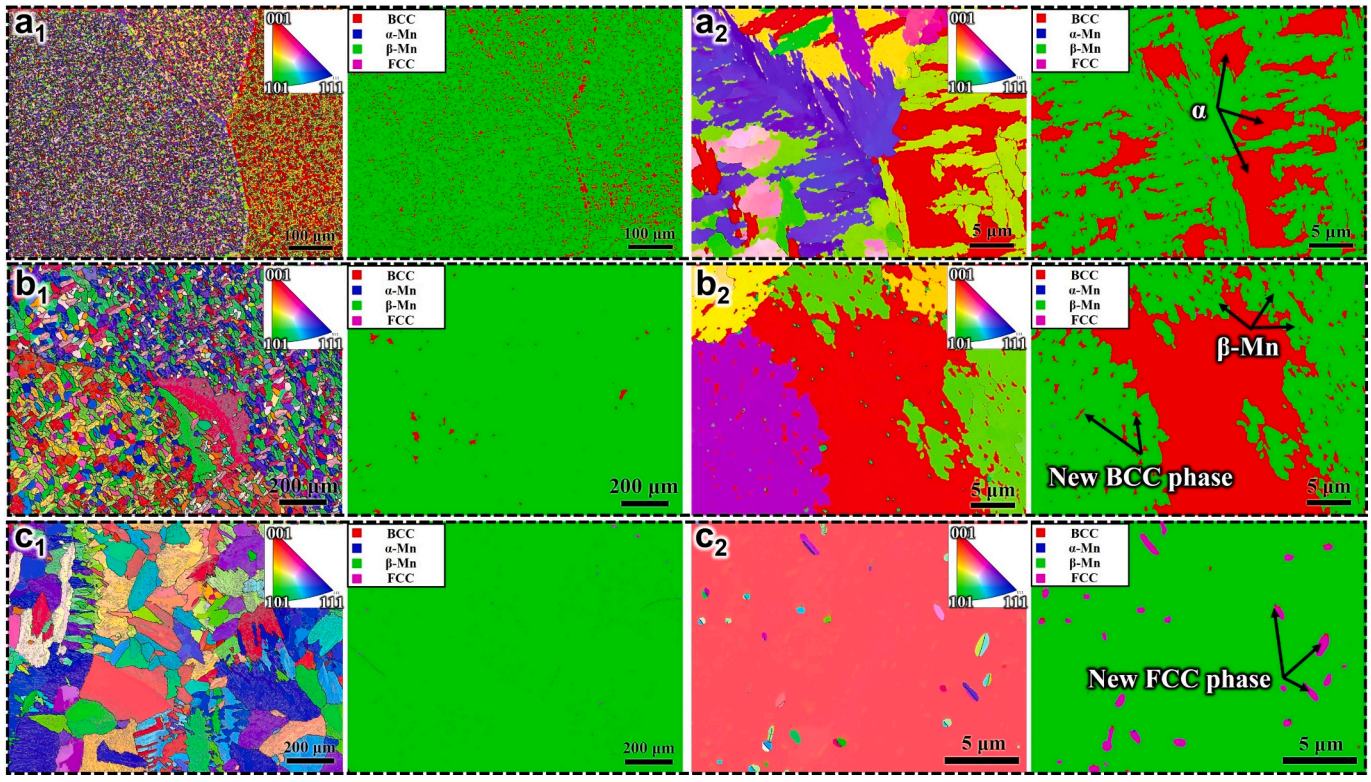


Fig. 4. EBSD inverse pole figures and corresponding phase maps of the 15Al alloy after aging at (a_1) 525 °C for 3 h, (b_1) 550 °C for 6 h, and (c_1) 650 °C for 3 h. (a_2-c_2) are the magnified views of (a_1-c_1).

presents the SE images of the 15Al alloy subjected to different aging treatments. Based on the previous study [15], the SE images can be used to identify the precipitation of secondary phases such as α-Mn and β-Mn phases in the aged samples. The α-Mn precipitates were dispersed in the α matrix as fine particles. The short rod-shaped β-Mn phase precipitated along α grain boundaries and the interior of α grains, and gradually grew until the precipitates came into contact with each other. Specifically, after short-time aging at low temperatures (e.g., aging at 525 °C for 1 h), a large amount of α-Mn densely precipitated in the α matrix, while the formation of β-Mn was minimal, resulting in only a slight increase in hardness from 253.4 HV to 379.8 HV. This indicates that the hardening effect of α-Mn is limited. However, the rate of precipitation and growth

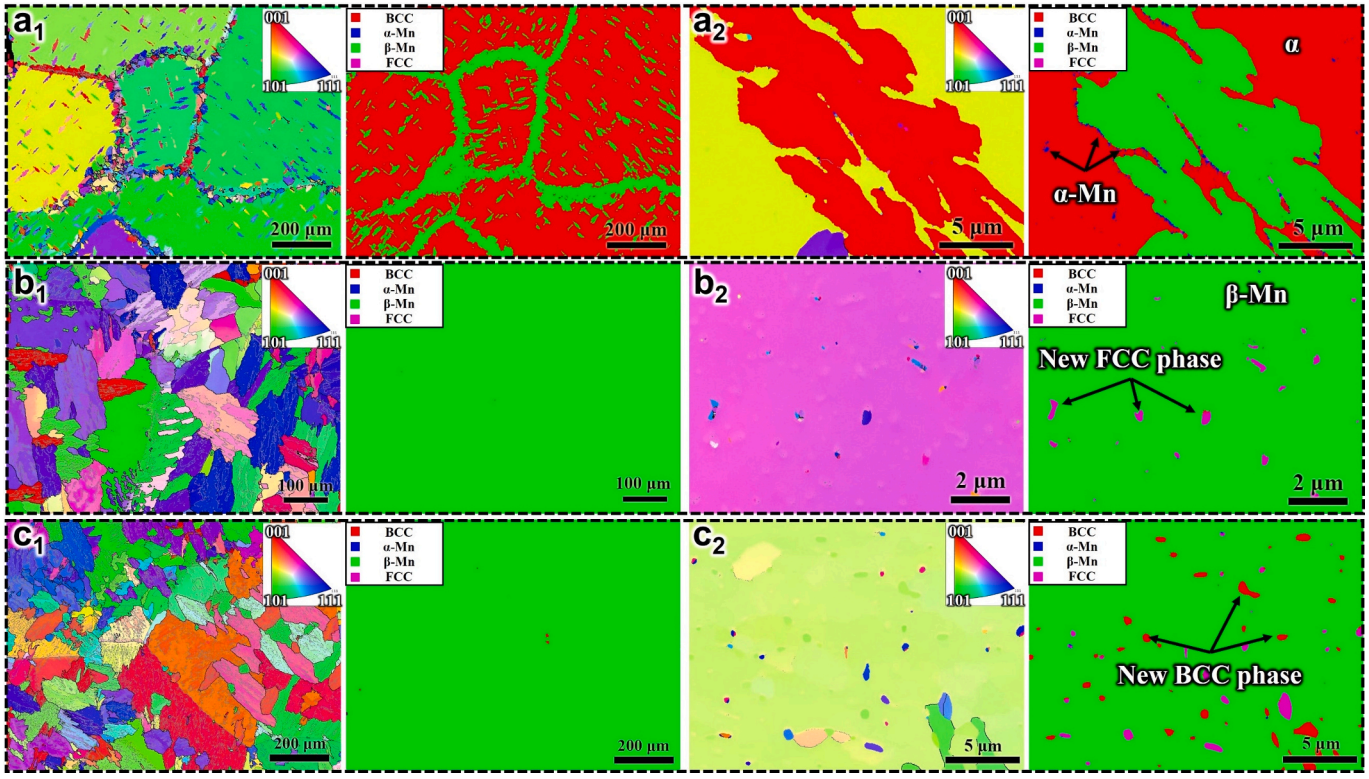


Fig. 5. EBSD inverse pole figures and phase maps of the 17Al alloy after aging at (a_1) 525 °C for 10 h, (b_1) 575 °C for 6 h and (c_1) 650 °C for 10 h. (a_2-c_2) are the EBSD results of the magnified views of (a_1-c_1).

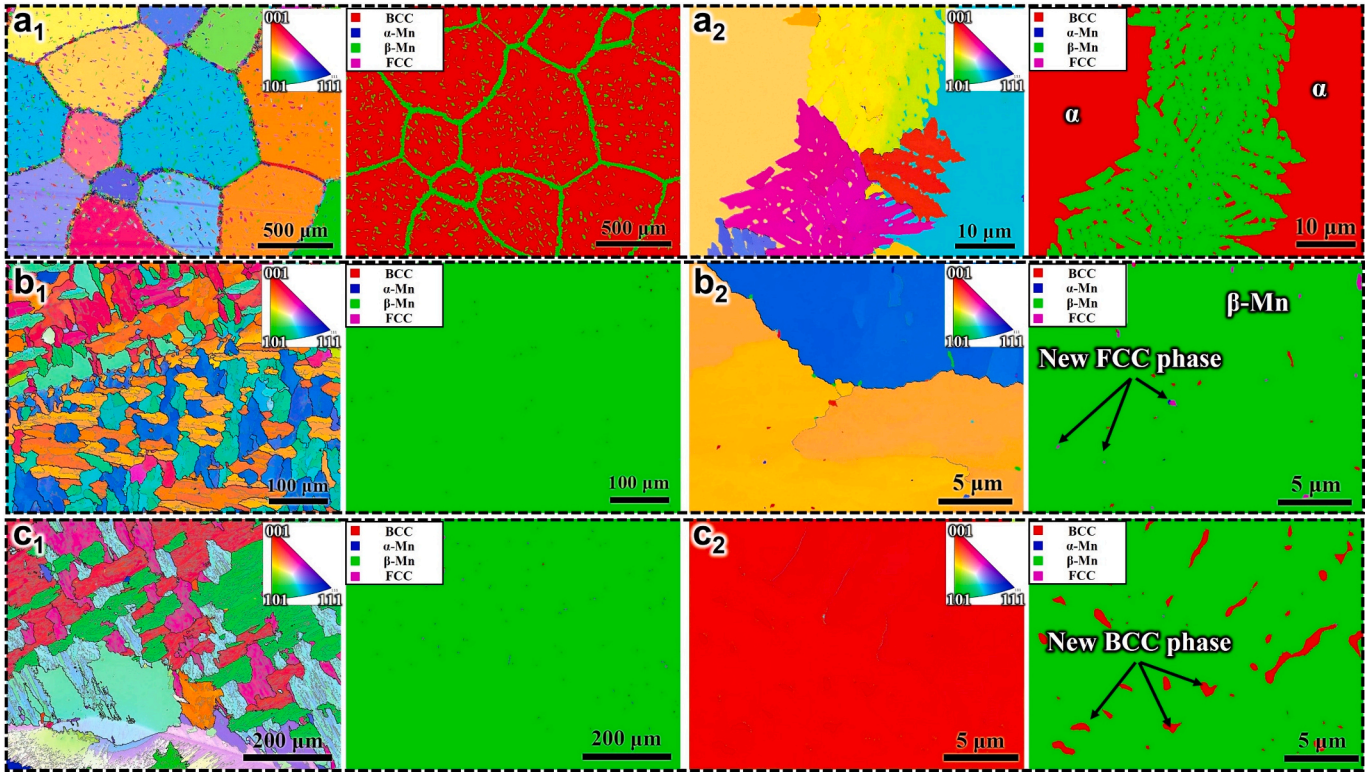


Fig. 6. EBSD inverse pole figures and phase maps of the 19Al alloy after aging at (a_1) 525 °C for 10 h, (b_1) 600 °C for 3 h, and (c_1) 650 °C for 6 h. (a_2-c_2) are the EBSD results of the magnified views of (a_1-c_1).

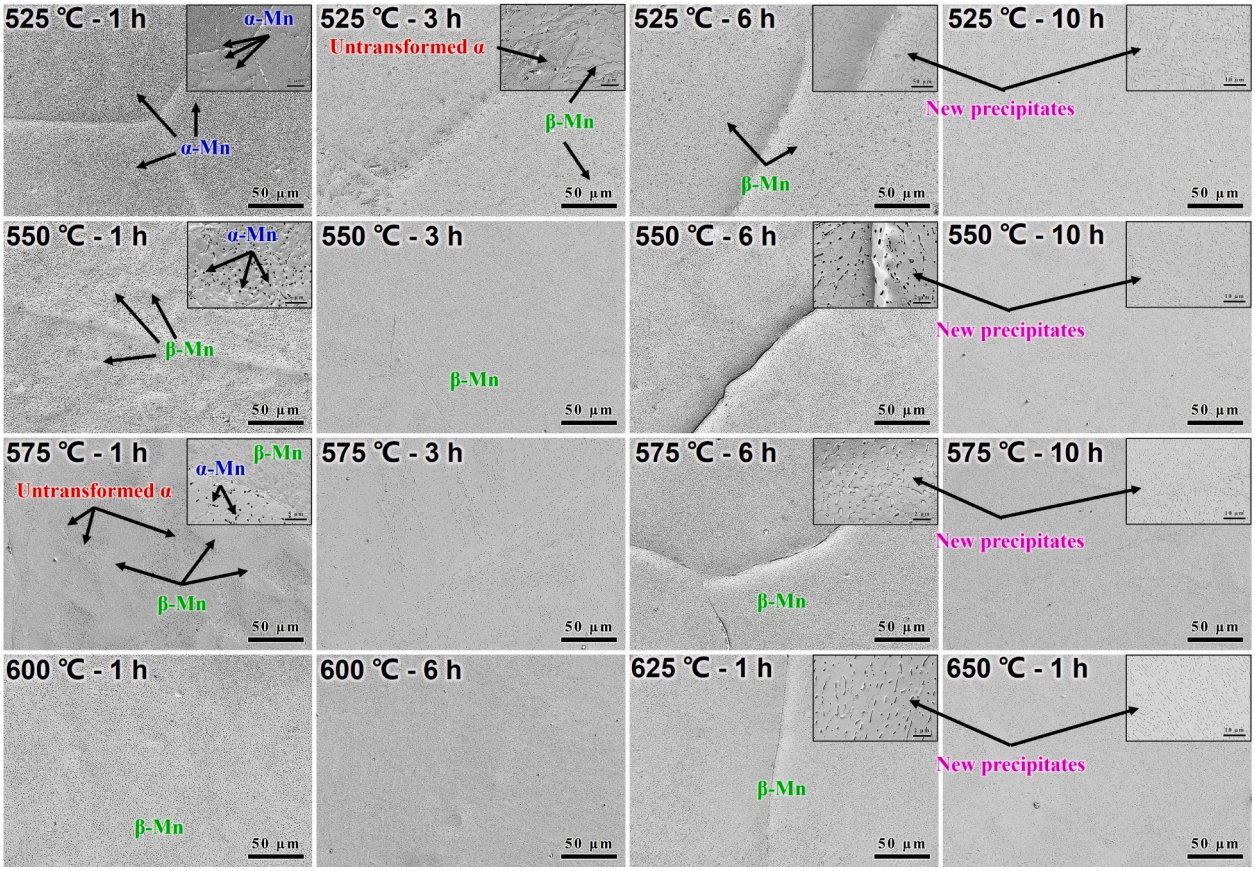


Fig. 7. Secondary electronic images of the 15Al alloy after different aging treatments.

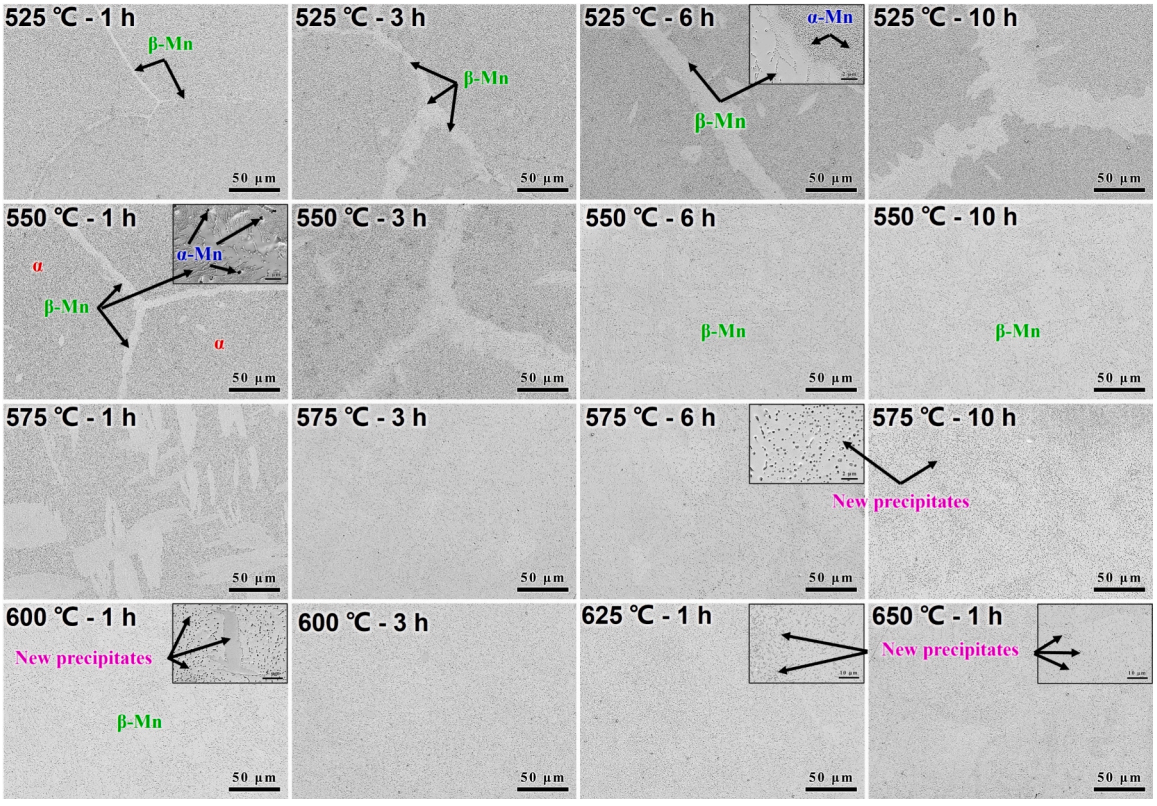


Fig. 8. Secondary electronic images of the 17Al alloy after different aging treatments.

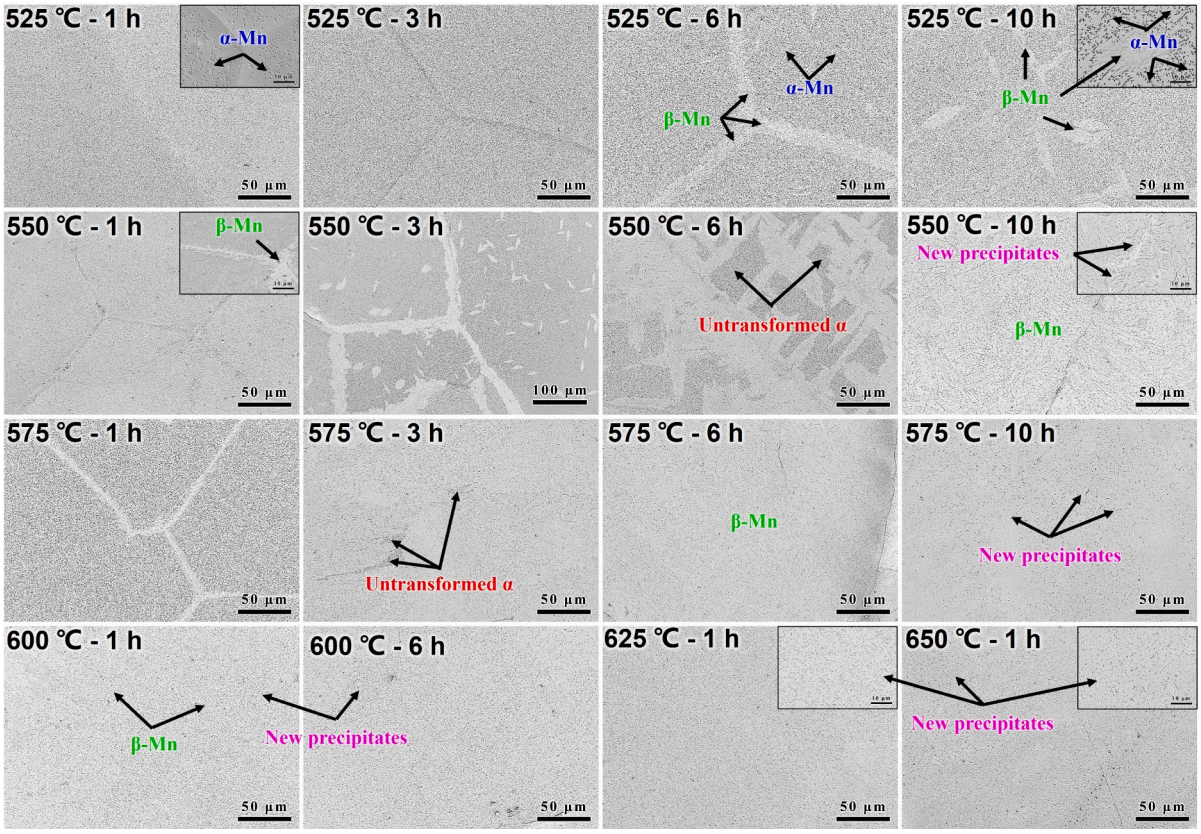


Fig. 9. Secondary electronic images of the 19Al alloy after different aging treatments.

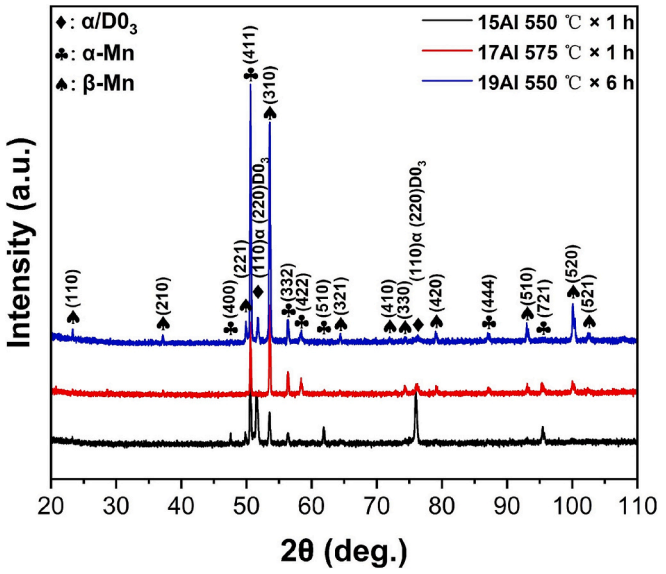


Fig. 10. The XRD results for the 15Al, 17Al and 19Al alloys subjected to different aging treatments.

for β-Mn was accelerated with increasing aging temperatures, partly attributing to the enhanced atomic diffusion rate. At 575 °C, a significant amount of β-Mn was formed after just 1 h of aging. The β-Mn with ultra-high intrinsic hardness led to the alloy exhibiting ultra-high hardness [26]. Consequently, the hardness of the alloy significantly increased. Due to the continuous precipitation of β-Mn with prolonged aging time, the α matrix would almost completely transform into β-Mn. When the alloy was aged at 600 °C and above, the α matrix had almost completely transformed into β-Mn within just 1 h. Furthermore, abundant fine precipitates were observed within the β-Mn. Combined with the EBSD results, it is evident that these precipitates were BCC and FCC phases.

Figs. 8 and 9 show the SE images of the 17Al and 19Al alloys after different aging treatments, respectively. It is evident that the increase in Al content slowed down the precipitation kinetics of β-Mn during the low-temperature aging processes. When 17Al and 19Al alloys were aged at 525 °C, the precipitation of β-Mn was very slow as compared to the 15Al alloy, and only a small amount of β-Mn was observed even after 10 h of aging. The β-Mn primarily precipitated along α grain boundaries, while a small amount of short rod-shaped β-Mn was also observed within the α grains. At 550 °C, the α matrix had nearly fully transformed into β-Mn in the 17Al alloy after 6 h of aging. As for the 19Al alloy, it required 10 h. After 1 h of aging at 575 °C, a small amount of β-Mn precipitated along α grain boundaries and within α grains in the 17Al alloy, while β-Mn in the 19Al alloy could only be observed along α grain boundaries. Even extending the aging time to 3 h, the untransformed α matrix was still observed in the 19Al alloy. However, at temperatures of 600 °C and above, the α had almost completely transformed into β-Mn in both alloys after just 1 h of aging. Numerous fine precipitates distributed within the β-Mn were also observed in the 17Al and 19Al alloys after the α matrix had fully transformed into β-Mn. Based on the SE results, the 15Al alloy aged at 550 °C for 1 h, 17Al alloy aged at 575 °C for 1 h, and 19Al alloy aged at 550 °C for 6 h were selected for XRD tests, and the results are shown in Fig. 10. Both α-Mn and β-Mn diffraction peaks were found in these XRD patterns. This indicates that α-Mn and β-Mn were co-precipitated during aging before the α matrix fully transformed to β-Mn.

3.3. TEM observations

TEM observations were conducted on the 15Al alloy aged at 600 °C for 6 h (965.6 HV) and the 17Al alloy aged at 575 °C for 6 h (924.9 HV)

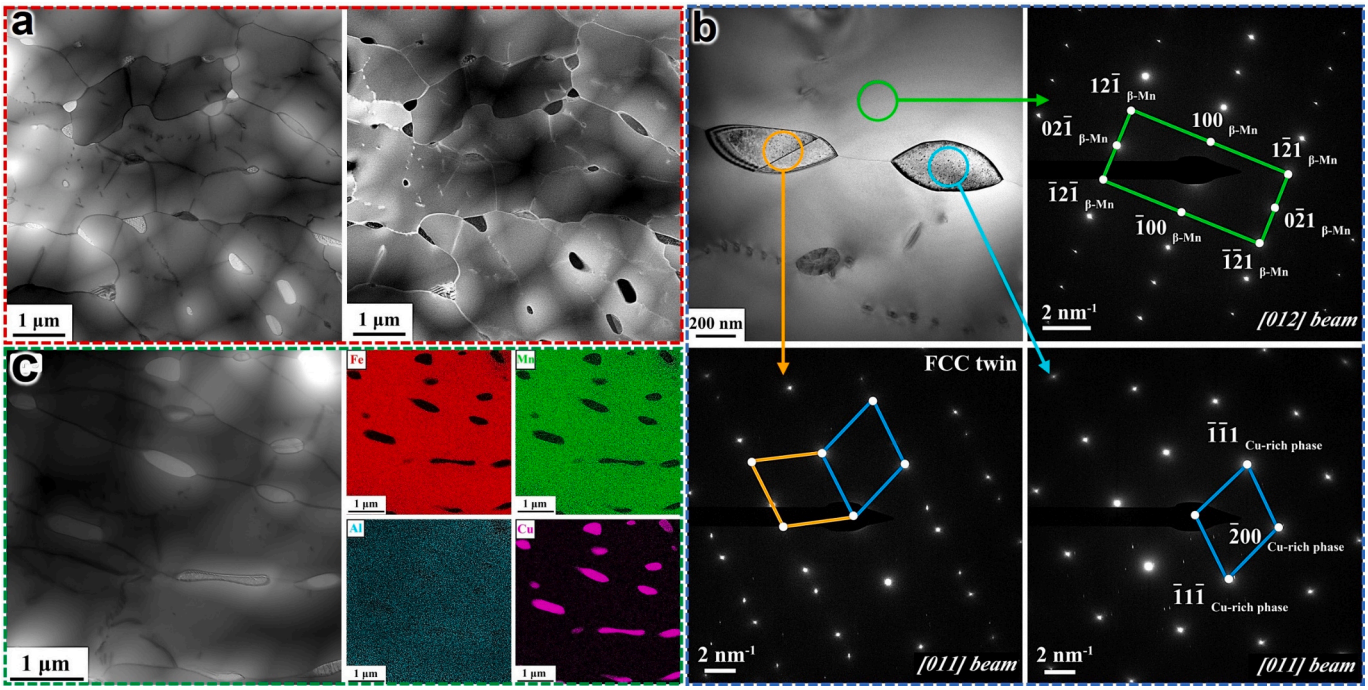


Fig. 11. TEM images of the 15Al alloy aged at 600 °C for 6 h. (a) Low-magnification bright-field (BF) image and corresponding dark-field (DF) image. (b) Selected area electron diffraction (SAED) patterns corresponding to the circles highlighted areas with different colors. (c) BF image and corresponding EDS elemental mapping.

to further elucidate the microstructures of these alloys with ultra-high hardness. As shown in Fig. 11 **(a, b)**, the α matrix completely disappeared and abundant near equiaxed β-Mn formed in the 15Al alloy after aging at 600 °C for 6 h. Besides, some fine particles are evenly distributed at the interfaces between the β-Mn. The selected area electron diffraction (SAED) patterns shown in Fig. 11 **(b)** indicate that the fine particles exhibited an FCC crystalline structure, with FCC twins frequently observed within the particles. This is consistent with the EBSD result shown in **Figs. 4, 5 and 6**. Fig. 11 **(c)** shows that the FCC phase was rich in Cu, depleted in Fe and Mn, and slightly poor in Al. Therefore, these precipitates were referred to as the FCC Cu-rich phase hereafter. In addition, the β-Mn was poor in Cu according to Fig. 11 **(c)**.

Fig. 12 presents the TEM results of the 17Al alloy after aging at 575 °C for 6 h. Similar to the results observed in Fig. 11 **(a)**, a large amount of fine particles were distributed at the interfaces of β-Mn (Fig. 12 **(a)**). EDS elemental mapping results indicated that these particles were the FCC Cu-rich phase, as shown in Fig. 12 **(b)**. FCC twin structure was also observed within the Cu-rich particles. Fig. 12 **(c)** provides the SAED pattern and high-resolution TEM (HRTEM) image of β-Mn. Fig. 12 **(d)** shows the HRTEM image of the interface between the β-Mn and FCC Cu-rich phase. Fast Fourier transform (FFT) analysis was conducted on different regions corresponding to the β-Mn and FCC Cu-rich phases to further confirm the presence of these two phases.

3.4. Intrinsic hardness of β-Mn

Fig. 13 presents the nanoindentation results for the hardest samples of the three Fe-Mn-Al-Cu low-density steels after aging treatments. The peak Vickers hardness values for the 15Al, 17Al, and 19Al alloys were 965.6 HV, 924.9 HV, and 917.8 HV, respectively. As shown in Fig. 13 **(a)**, the average hardness values of the same samples measured by nanoindentation were 11.7 ± 0.4 GPa, 11.1 ± 0.6 GPa, and 10.4 ± 1.3 GPa, respectively. Both the peak Vickers hardness and micro-hardness decreased with increasing Al content. The presence of the FCC Cu-rich and BCC phases resulted in variations in micro-hardness values across different areas, as shown in Fig. 13 **(b)**. The micro-hardness decreased when the indenter was pressed into the FCC Cu-rich and BCC phases, indicating that the intrinsic hardness of these precipitates were all lower than that of the β-Mn. The error in micro-hardness increased with increasing Al content, which laterally indicates that the content of these FCC and BCC precipitates also increased with the increase in Al content. We selected the areas where the nanoindentation indents did not contact the FCC Cu-rich and BCC phases to calculate the intrinsic hardness of the β-Mn. Accordingly, the β-Mn in hardest samples of 15Al, 17Al, and 19Al alloys showed the average intrinsic hardnesses of 12.1 ± 0.3 GPa, 12.2 ± 0.1 GPa, and 12.5 ± 0.4 GPa, respectively. Because of the high intrinsic hardness of β-Mn, the alloys exhibited rapid hardening response during the aging treatments and achieved ultra-high hardness. Additionally, the chemical compositions of β-Mn in the three samples were measured by SME-EDS, and the results are shown in Fig. 14. In the hardest samples of 15Al, 17Al, and 19Al alloys, the Al contents in the β-Mn were 14.3 at.%, 16.3 at.%, and 17.5 at.%, respectively. However, the contents of other elements in the three alloys were very close to each other. These results indicate that the increase in Al content slightly enhanced the intrinsic hardness of β-Mn in 15Al, 17Al, and 19Al alloys through solid-solution strengthening. However, as can be seen in Fig. 13 **(b)**, the FCC Cu-rich and BCC phases between the β-Mn interfaces gradually increased with increasing Al content. Since the intrinsic hardness of these precipitates was lower than the intrinsic hardness of β-Mn, in this case, it resulted in the overall hardness showing a pattern of decreasing with increasing Al content.

Since the ultra-high hardness would result in a decrease in the toughness, cracks were generated during the Vickers hardness testing. The average crack lengths resulting from the hardness measurements were 18 μm for 15Al alloy, 22 μm for 17Al alloy, and 35 μm for 19Al alloy. Therefore, the calculated K_{IC} values by using Eq. (1) for the hardest samples of the 15Al, 17Al, and 19Al alloys were 8.6 MPa·m$^{1/2}$, 6.6 MPa·m$^{1/2}$, and 3.5 MPa·m$^{1/2}$, respectively. This indicates that the K_{IC} of ultra-hard Fe-Mn-Al-Cu low-density steels decreased with increasing Al content.

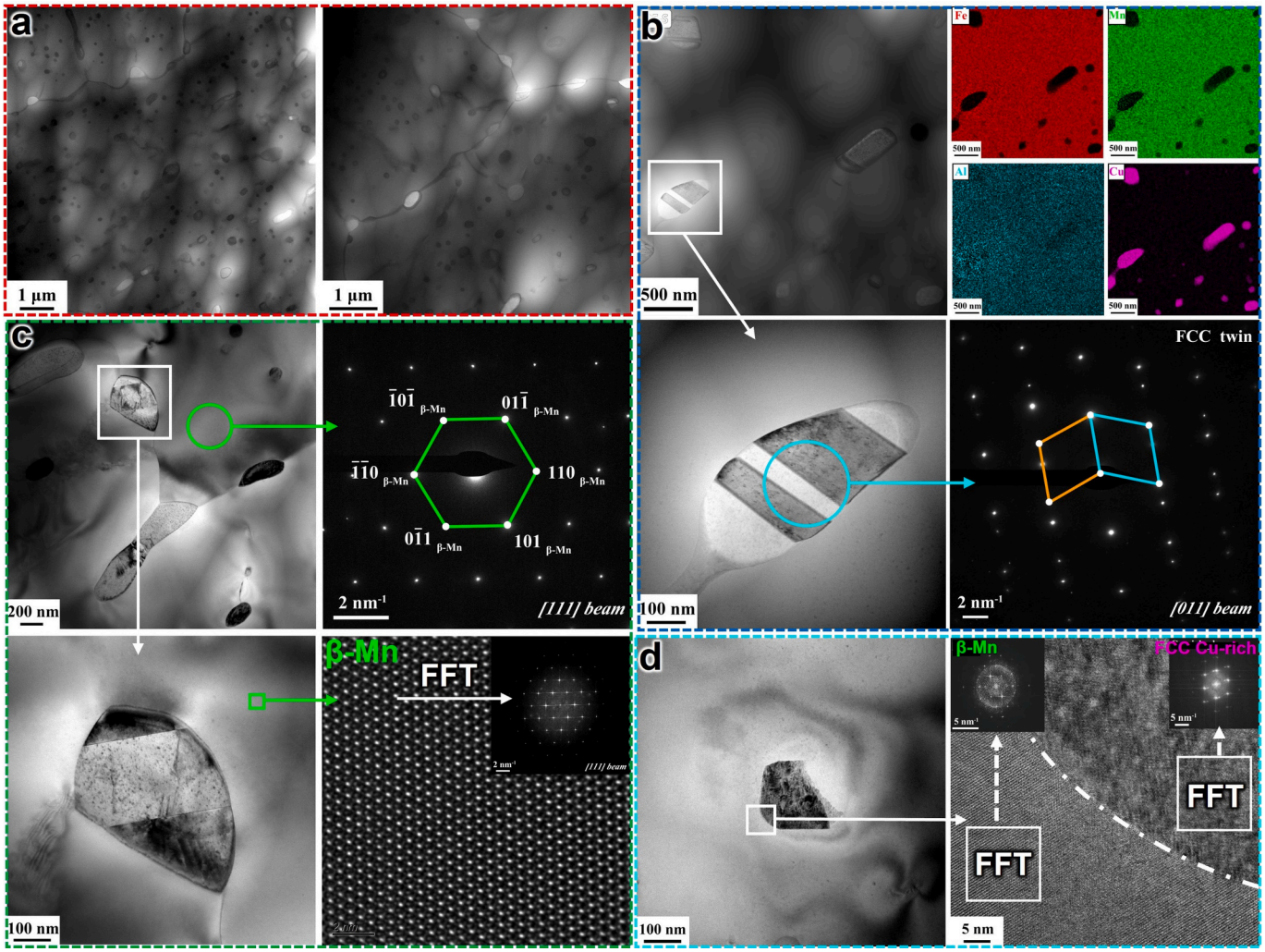


Fig. 12. TEM images of the 17Al alloy aged at 575 °C for 6 h. (a) Low-magnification BF images. (b) High-magnification BF image and corresponding EDS elemental mapping, as well as the SAED pattern corresponding to the area highlighted by the blue circle (FCC twins). (c) BF images and corresponding SAED and high-resolution TEM (HRTEM) image of the β-Mn regions. (d) HRTEM image showing the interface of the Cu-rich phase and β-Mn. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4. Discussions

4.1. Effect of Al contents on aging hardening behaviors of Fe-Mn-Al-Cu low-density steels

The combined microstructure observations and nanoindentation results indicated that the key to achieving ultra-high hardness in Fe-Mn-Al-Cu low-density steels through aging lies in the complete phase transformation from the α matrix to β-Mn. In this study, we investigated the effects of Al contents on the aging hardening responses of Fe-Mn-Al-Cu low-density steels. The results showed that the increase in Al contents decreased the precipitation kinetics of β-Mn. When the Al content was 15.0 at.%, β-Mn precipitated rapidly during the aging processes at temperatures of 525 °C and above. For example, within 3 h of aging at 525 °C, the α matrix had been almost completely transformed into β-Mn (Fig. 7). However, for the 17Al and 19Al alloys, this transformation was only completed after 6 h and 10 h when aging at 550 °C, respectively (Figs. 8 and 9). Both the precipitation and growth of β-Mn involve the process of atomic diffusion. The decrease in the precipitation kinetics of β-Mn with increasing Al contents is explained as follows: 1) The β-Mn is a Mn-rich phase, and its nucleation and growth must exclude large amounts of other atoms to the surroundings. As shown in Fig. 15, the β-Mn is rich in Mn while poor in Al and Cu. In this case, the increase in Al content resulted in the enrichment of more Al atoms around β-Mn, which would hinder its further precipitation and growth. 2) In Fe-Mn-Al-based alloys with high Al contents, the precipitation of the $D0_3$ phase after solution treatment and quenching is quite common [30,31]. The increase in Al content will lead to an increase in the amount and size of the $D0_3$ phase in Fe-Mn-Al-based alloys, attributing to the strong stabilizing effect of the Al element on the Al-rich $D0_3$ phase [32,33]. In our previous study, the precipitation of the $D0_3$ phase in α-matrix was observed in both Fe-37.8Mn-6.9Al and Fe-36.5Mn-6.5Al-7.7Cu (wt.%) alloys after solution treatment. Moreover, the $D0_3$ phase would continue to precipitate and grow during the subsequent aging treatment [15]. However, the $D0_3$ phase was not observed in alloys with the maximum hardness, which suggests that the precipitation and growth of β-Mn were accompanied by the depletion of the $D0_3$ phase. Since the increase in Al content would lead to an increase in the amount and size of the $D0_3$ phase and also stabilize it, this might hinder the the nucleation and growth of β-Mn due to the prerequisite of the depletion of the $D0_3$ phase. The increase in aging temperature accelerated the precipitation and growth of β-Mn, a phenomenon attributed to the enhanced driving force for β-Mn formation and the accelerated atomic diffusion rate [15]. The increase in the volume fraction of β-Mn significantly enhanced the hardness, as the β-Mn exhibited an intrinsic high hardness of over 10 GPa [24].

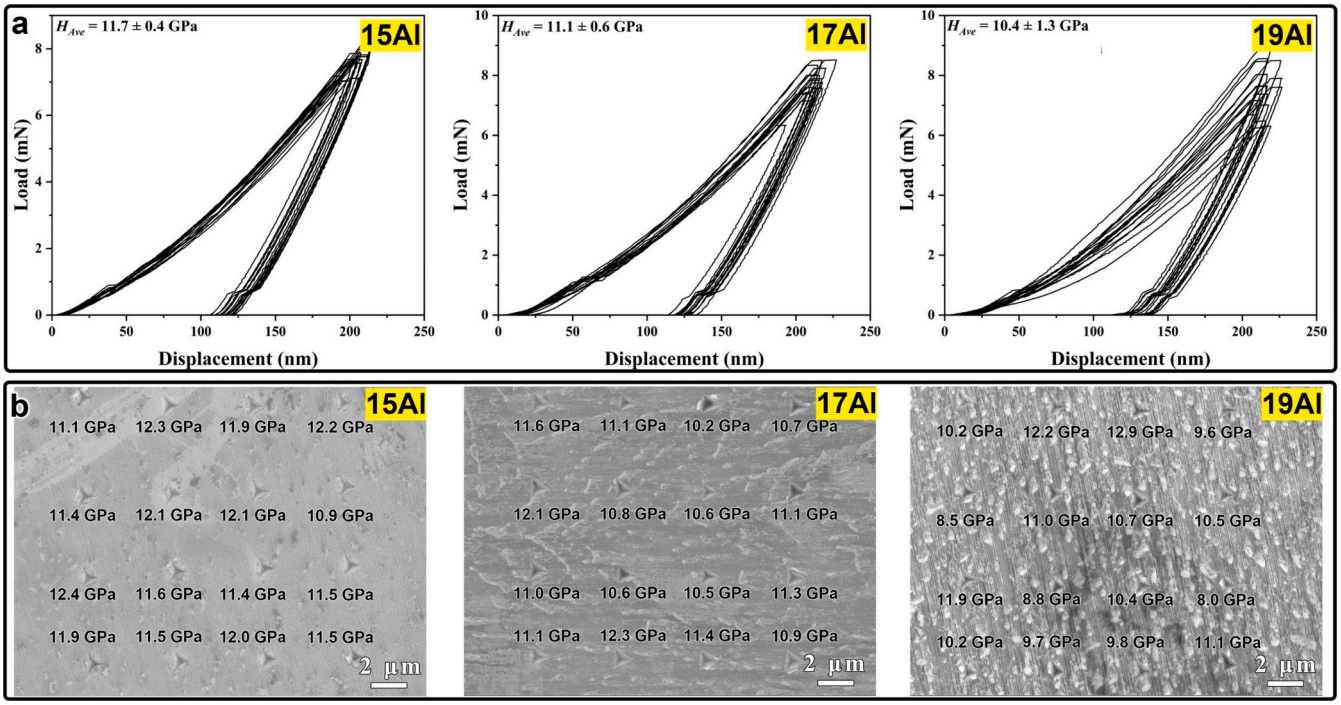


Fig. 13. Nanoindentation results of the 15Al alloy aging at 600 °C for 6 h, 17Al alloy aging at 575 °C for 6 h, and 19Al alloy aging at 600 °C for 10 h. (a) Nanoindentation curves. (c) SE images with corresponding EDS elemental mapping of indentations. (b) Overall SE image of the indentations.

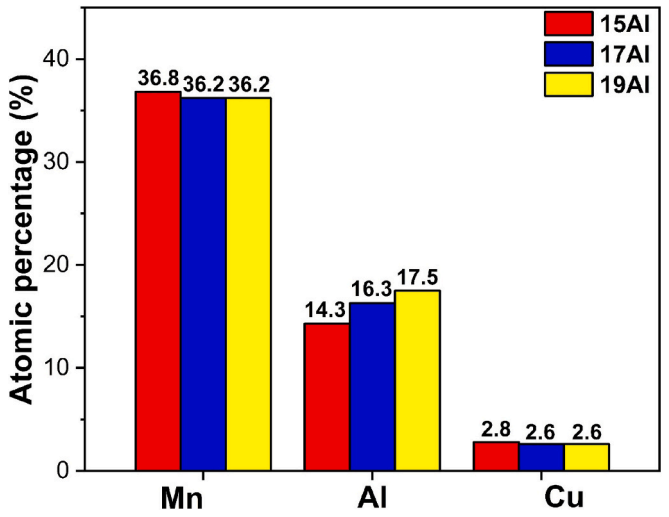


Fig. 14. SEM-EDS results for β-Mn in the hardest samples of the 15Al, 17Al, and 19Al alloys.

The contents of Al also exerted a significant influence on the microstructures of ultra-hard alloys. In the alloys with peak hardness value, new BCC phase and FCC Cu-rich phase formed at the interfaces between the β-Mn. Although only the FCC Cu-rich phase was characterized in the TEM results (Figs. 11 and 12), the EBSD results (Figs. 4, 5, and 6) clearly revealed the presence of the BCC phase within the β-Mn. Meanwhile, the results of our previous study also confirmed the existence of the BCC phase [15]. It is evident that the size distribution of the BCC phase particles is comparable to that of the FCC Cu-rich phase. As the β-Mn is an Mn-rich phase, its nucleation and growth involve the diffusion and redistribution of elements, which enriched Mn while expelling Fe, Al, and Cu into the surrounding matrix. Therefore, the formation of the BCC phase at the interfaces between the β-Mn is attributed to the enrichment of BCC stabilizing element Al around the interfaces of β-Mn [25]. The Cu-rich phase was formed due to the limited solubility of Cu in Fe-based alloys and the expelling of Cu from the β-Mn during its precipitation and growth [15,34]. The microstructure evolutions during aging treatments of the Fe-Mn-Al-Cu low-density steels followed this sequence: α → α + FCC Cu-rich phase + α-Mn + β-Mn + BCC phase→ β-Mn + BCC phase + FCC Cu-rich phase. The precipitates between β-Mn interfaces in the ultra-hard alloys gradually increase with the increase in Al content (Fig. 13). Furthermore, the EBSD results (Figs. 4, 5 and 6) show that the content of the FCC Cu-rich phase between β-Mn interfaces decreased gradually with the increase of the Al content.

According to the nanoindentation results, the increase in Al content would lead to the decrease in the micro-hardness of ultra-hard Fe-Mn-Al-Cu low-density steels (Fig. 13). Although the addition of Al enhanced the intrinsic hardness of the β-Mn through solid-solution strengthening, this enhancement was minimal as reflected by a marginal increase in the intrinsic hardness of the β-Mn [29]. Besides, the increase in Al content led to the increase of the content of FCC Cu-rich phase and BCC phase between the β-Mn interfaces. It is evident that both the intrinsic hardnesses of FCC Cu-rich and BCC phase were all significantly lower than that of β-Mn. The ultra-hard Fe-Mn-Al-Cu low-density steels were composed of β-Mn, as well as the FCC Cu-rich and BCC phases distributed between their interfaces. As a result, the hardness depended not only on the intrinsic hardness of β-Mn, but also on the absolute volume fraction of β-Mn. Elevated Al content led to the increase of FCC Cu-rich phase and BCC phase between the β-Mn interfaces, resulting in the decrease in the absolute volume fraction of β-Mn. Taking the EBSD results of the three alloys aged at 650 °C as an example, the volume fraction of FCC Cu-rich and BCC phases of the 15Al alloy after aging at 650 °C for 3 h was 3.9 %, and the Vickers hardness of this sample was 852.3 HV. The volume fraction of these precipitates of the 17Al alloy after aging at 650 °C for 10 h was 4.7 %, and the average Vickers hardness of the alloy was 833.2 HV. However, after the 19Al alloy aged at 650 °C for 6 h, the volume fraction of FCC Cu-rich and BCC precipitates reached 6.5 %, and its Vickers hardness was only 810.9 HV. Ultimately, this led to the decrease in the overall micro-hardness as well

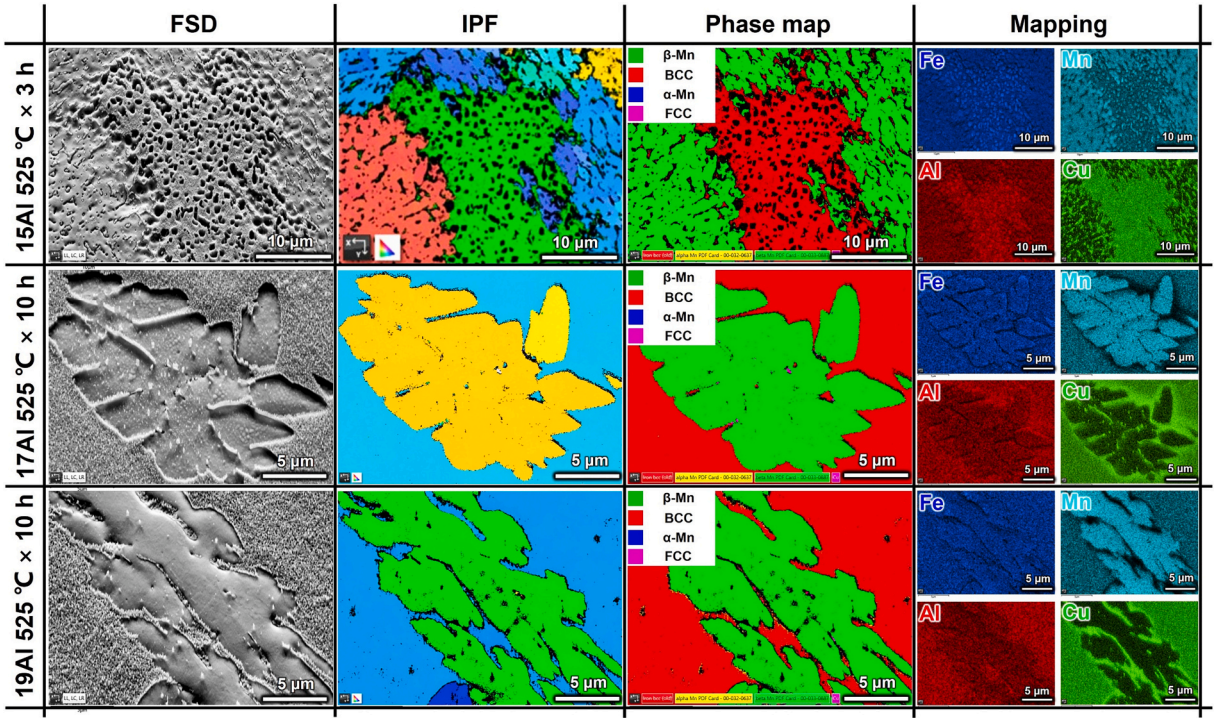


Fig. 15. EBSD Forescatter detector (FSD) images, inverse pole figures (IPF), phase maps, and EDS mapping for the 15Al (525 °C for 3 h), 17Al (525 °C for 10 h) and 19Al (525 °C for 10 h) alloys.

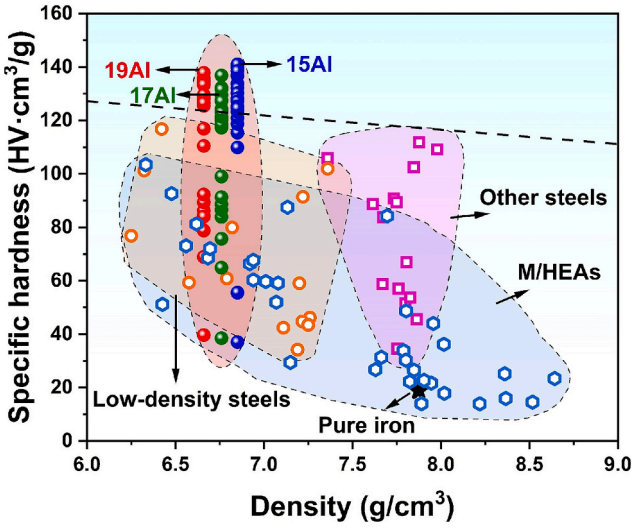


Fig. 16. Specific hardness (hardness/density, HV/ρ) versus density among our Fe-Mn-Al-Cu low-density steels and various low-density steels, other steels (e.g., stainless steels, high-speed steels, ferritic/martensitic steels, carbon steels, etc.), pure iron, and medium/high entropy alloys.

as the highest Vickers hardness value of Fe-Mn-Al-Cu low-density steels with increasing Al content. In addition, the K_{IC} of Fe-Mn-Al-Cu low-density steels decreased with increasing Al content. Firstly, the intrinsic hardness of β-Mn gradually increased with increasing Al content, and the K_{IC} should be deteriorating accordingly. Moreover, the FCC phase exhibited relatively low intrinsic hardness but is more ductile, while the BCC phase is usually harder than the FCC phase [35]. Therefore, the decrease in ductile FCC Cu-rich phase due to increasing Al content led to the decrease in fracture toughness of ultra-hard alloys. The above two reasons together contributed to the decrease in the K_{IC} of Fe-Mn-Al-Cu low-density steels with increasing Al content.

4.2. Insights into Fe-Mn-Al-Cu low-density steels

In this study, we achieved exceptionally wide adjustable hardness values, ranging from 253.4 HV to 965.6 HV in the Fe-Mn-Al-Cu low-density steels through simple aging treatments. Notably, different hardness levels were obtained in these alloys by controlling aging parameters, thereby meeting various practical application requirements. Fig. 16 compares the specific hardness (hardness/density, HV/ρ) of our Fe-Mn-Al-Cu low-density steels with various low-density steels, other types of steels, pure iron, and medium/high entropy alloys (M/HEAs) [17,18,35–73]. The Fe-Mn-Al-Cu low-density steels not only exhibit ultra-high hardness across a wide range of flexibility but also boast the advantage of low density. The hardness of existing steels and M/HEAs generally does not exceed 800 HV and coupled with the higher densities. This results in their specific hardness usually below 120 HV·cm^3/g. In contrast, our Fe-Mn-Al-Cu low-density steels achieve a maximum specific hardness exceeding 140 HV·cm^3/g while maintaining low density. Furthermore, the Fe-Mn-Al-Cu low-density steels with ultra-high hardness also demonstrate acceptable fracture toughness. For instance, the 15Al alloy maintains a fracture toughness of 8.6 MPa·m$^{1/2}$ even at a high hardness level of 965.5 HV. This represents a significant breakthrough in the research on high-performance low-density steels.

Ultra-high hardness over 950 HV was achieved in Fe-Mn-Al-Cu low-density steels with high Al contents. After the simple short-time aging treatments, the continuous precipitation of β-Mn led to the achievement of an ultra-wide range of flexible and tunable hardness. This means that these alloys can be formed into parts or components with different shapes and sizes in the soft state (solution-treated) and then aged to obtain the appropriate hardness range, which could significantly reduce the preparation cost and difficulty of traditional high hardness and wear-resistant components. In the alloys studied in this work, the increase in Al content would slow down the precipitation rate of β-Mn and decrease the fracture toughness of the alloys with peak hardness. Therefore, the change in the mechanical performance of Fe-Mn-Al-based low-density steels needs to be considered comprehensively when reducing their density. In summary, our Fe-Mn-Al-Cu alloys are able to

increase the hardness dramatically within short-time aging, saving fabrication time and costs. In addition, its low density and flexible hardness can realize “one type of material, multiple uses” under the condition of meeting the application premise of lightweight structural materials. Despite the superior specific hardness of these Fe-Mn-Al-Cu alloys, the toughness remains not good enough. Therefore, achieving an optimal combination between hardness and toughness warrants further study.

5. Conclusions

In this work, we systematically investigated the effects of Al contents on the aging hardening behaviors of the Fe-Mn-Al-Cu low-density steels subjected to different aging treatments and derived the following conclusions:

1) An ultra-wide flexible hardness ranging from 253.4 HV to 965.6 HV was tailored in Fe-Mn-Al-Cu low-density steels with different Al contents. Due to the combination of ultra-high hardness and low-density advantages, the Fe-35.2Mn-15.0Al-5.2Cu alloy achieved an ultrahigh specific hardness of 141 HV·cm^3/g.
2) The increase in Al contents slowed down the precipitation kinetics of β-Mn during aging treatments, which led to the complete transformation from α to β-Mn being extended.
3) The increase in Al content slightly enhanced the intrinsic hardness of β-Mn. However, it also led to the decrease of FCC Cu-rich phase between β-Mn interfaces and the decrease in the fracture toughness of the ultra-hard alloys. This resulted in the decrease in the final hardening effect of Fe-Mn-Al-Cu low-density steels with increasing Al content.

CRediT authorship contribution statement

Dongwei Wang: Writing – original draft, Investigation. **Yang Zuo:** Writing – original draft, Methodology, Investigation. **Lixin Sun:** Methodology, Investigation. **Fanyu Wang:** Methodology. **Jiankun Xu:** Methodology. **Yang Zhao:** Methodology. **Yu Fu:** Writing – review & editing, Supervision. **Huabei Peng:** Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time due to technical or time limitations.

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